

This is where the openness of Android kicked in. Soon, developers and hackers were busy trying to circumvent the restrictions to fully utilize the potential of their smartphones. This led to a number of tweaks and after-market ‘firmwares’ being made available to Android, a feature very unique (and central) to the Android ecosystem. Soon regular users were able to tweak the software, it’s underlying components and even change the way the hardware worked using some relatively simple techniques. This led to concepts like ‘rooting a phone’ and installing a ‘Custom ROM’ - terms any Android enthusiast will be very familiar with. It is worth mentioning at this point that Android offers far more flexibility and customisability compared to rival platforms like iOS, Blackberry OS or Windows Phone. While other OSs only skim the surface in this regard, Android lets you alter the look and feel of the device without having to resort to a root or an alternative firmware. Themes, icon packs, home screen replacement apps and widgets for nearly every plausible feature will keep you in control of your device to a fair extent. Android also allows users to install apps not found in the official Google Play Store (called ‘sideloading’ apps), which many others like iOS and Windows Phone don’t.

In this chapter, we give you lowdown on what it’s like to unleash the full power of your Android device. While it’s nearly impossible to give a detailed guide for rooting or flashing a ROM for every device out there (due to the sheer number and variety of different /Android devices) we’ll cover some essentials concepts along with their benefits. Bear in mind though, that while they may unlock many new features on your device, they do involve some risk. At the very least you would be voiding your device’s warranty, or worst-case you might end up ‘bricking’ your phone, rendering it completely unusable. On that note, let’s dive in to the wonderful world of Android!

## Rooting your device

As discussed earlier, even though Android is supposed to be Open Source, users don’t always have access to all of the phone’s features or can’t modify the phone in an arbitrary way. One of the (if not the most) popular starting points for Android customization is to ‘root’ your phone. Rooting gives the user access to a plethora of tweaks and powerful customizations include the ability to tether your phone’s internet connection via Wi-Fi, overclocking your phone’s CPU, backing up your apps or installing ad-blockers. But what exactly is rooting and what do you, as a user stand to gain from the process? Let’s find out!